

TABLE III
ARDUINO FOOTPRINT-COMPATIBLE BOARDS

Name	Processor	Maker
SaintSmart Mega 2560	ATmega2560	SainSmart 
Freeduino SB	ATmega328	Solarbotics Ltd 

TABLE IV
SPECIAL-PURPOSE ARDUINO-COMPATIBLE BOARDS

Name	Processor	Maker
ArduPilot	—	
WIOT (Wireless Internet of Things)	—	

TABLE V
SOFTWARE-COMPATIBLE ONLY

Name	Processor	Maker
Boarduino	ATmega168 or ATmega328	Adafruit 

TABLE VI
NON-ATMEGA BOARDS

Name	Processor	Maker
LeafLabs Maple	STM32 (Cortex-M3)	LeafLabs 
Freescale Freedom	Kinetis-L (Cortex-M0+)	Freescale 
Sakura	Renesas RX63N	Renesas 
GR-KAEDE	RX64M	Renesas 



Fig. 2: The world's cheapest computer, CHIP

The long-term success of an SBC relies heavily on the performance-price ratio. The best performance-cost ratio comes from ODROID-U2, ODROID-X2, Raspberry Pi model B +, UG802, MK802, Cubieboard, VIA APC and Hackberry Board. The best RAM-cost ratio can be found in ODROID-U2, Hackberry Board and Raspberry Pi model B +.

Arduino-compatible boards

Although hardware and software designs are freely available under copy-left licences, developers have requested that the name Arduino be exclusive to the official product and not be used for derivative works without permission. As a result, a group of Arduino users forked Arduino Diecimila, releasing an equivalent board called Freeduino. Tables III, IV, V and VI enlist some common Arduino-compatible boards.

The future of development boards

Many of today's SBCs have the capability of modern-day PCs and tablets. This trend will continue as more powerful processors make their way into the embedded computing market as ever-increasing performance-price ratios rise, as well as additional manufacturers enter into this wild-west frontier of supporting open source hardware and software for both DIYers and professionals alike.

An additional trend we will continue to see is the availability of more accessories or add-on boards to be added to current SBC plat-

forms, giving users more options to control and have access to the outside world.

Professional engineers, on the other hand, can take these accessories and quickly add additional functionalities to their SBCs to develop working prototypes for currently on-hand projects.

CHIP. The world's cheapest computer, costing US\$ 9, is slimmer than a credit card. It is a tiny chip that can be used for editing documents, browsing the Internet and playing games. If used rightly, CHIP is capable of powering a wide range of small connected devices where cost-effectiveness and portability are of prime importance. It offers limited features when compared to other advanced computers, but considering the price, it is a very good buy.

System on module (SoM)/computer on module (CoM). This is a more appropriate choice for building embedded products. It is a complete embedded computer built on a single circuit board. With Moore's Law in action, microprocessors and memories have moved to smaller process nodes within two years, which is less than the expected standard product life time.

Customers' demands, which include advanced performance, low power consumption and compact machines, necessitate frequent design updates to the embedded platform. These design updates add to the development cost. By using pin-compatible SoMs for embedded development, we can escape from this deficiency.

Pin-compatible modules from Toradex enable plug-and-play for scaling-up platforms based on future technologies and market requirements. New modules can be easily connected to existing carrier boards; the application software may need some minor updates. **EFY**